

POWERBOX · PROBLEM STATEMENT

PowerBox problem statement

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/powerbox		
Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (users, context, constraints, out of scope, prior work)
0.1	2026-09-24	Amish Chadha	Initial scaffold

PowerBox problem statement

When the power goes out, households need a small, safe store of electricity for lights, phones, radio and internet, charged from whatever source is available: human power, a solar panel or the grid when it is up. Design with, not for: requirements must come from co-design sessions and field trials with the intended users through a local partner.

■ The problem

Outages are routine for a large share of the world. Grids with scheduled load shedding, storm-exposed distribution lines and informal connections leave households dark for hours, often in the evening when lighting and phone charging matter most. The fallback is usually a kerosene lamp, candles, a car battery on an unsafe charger, or a petrol generator, which is noisy, costly to run and a carbon monoxide hazard indoors.

Commercial portable power stations solve part of this, but they have three gaps for these users:

- **Closed, sealed batteries.** When the cells wear out, the whole unit is usually discarded. The pack cannot be swapped, shared or repaired.
- **Charging assumes a working grid or a large solar array.** Few accept human power, and none are designed to share a battery with an e-bike or a village charging hub.
- **Overselling.** Marketing implies a box can run a fridge or kettle. A household that buys on that promise and then drains the battery in an hour loses trust in the whole category.

PowerBox is a standalone power station built around the portfolio's SwapCell pack, so the same battery can move between a PowerBox, a SunSpoke e-bike and a SwapCell dock. It charges from StepGen (a manual stepper generator), a solar panel, the grid or a charged pack brought home from the bike, and it is honest about scale: one hour of steady human effort stores about 60 to 100 Wh, and one pack runs lights, phones, a radio and a router through an outage, not a fridge or a kettle.

■ Users and context

User	Need	Context
Outage-prone urban and peri-urban household	Evening light, phone charging, radio and home internet through a 2 to 6 h outage, recharged from the grid when it returns	Load shedding or storm outages; small homes and flats; indoor use
Off-grid rural household	A daily store for lights and phones, charged from a small solar panel or human power; a pack that can travel on an e-bike to a charging hub	No grid, or an unreliable mini-grid; dust, heat, rough transport
Household in an informal settlement	Safe light and phone charging without unsafe improvised wiring or kerosene; low purchase cost; theft-resistant design	Dense housing, fire risk, shared walls, limited space
Community charging point operator	Charge many phones and lamps for a small fee; swap packs between boxes and a charging hub; see how much energy each pack delivered	Market stall, school, clinic waiting room, community centre
Local technician or repair shop	Open design, standard parts, a pack and modules that can be diagnosed and replaced	Informal electronics repair trade
Open hardware community	A reference design that ties SwapCell, StepGen and SunSpoke together	Makerspaces, university labs, NGOs

■ Constraints

- Garage-buildable prototype. Concept budget about \$450 USD for the PowerBox parts; see the requirements for whether the SwapCell pack is included.
- Built around the SwapCell interface (v0.2 draft): 13S lithium-ion pack, about 46.8 V nominal (39.0 to 54.6 V), about 468 Wh, 340 x 90 x 80 mm body, blind-mate connector with CAN. PowerBox must not change the interface locally; conflicts go back to SwapCell.
- Standalone only. PowerBox must never connect to household wiring, a wall outlet or a distribution board. Back-feeding endangers line workers and is illegal almost everywhere.
- Carryable by one adult: a case with a single handle, small enough for a shelf or under a bed.
- Safe indoors: lithium cells, a mains-voltage AC outlet and hot power electronics all sit in a living space, often near children.
- Off-the-shelf modules where possible (certified charger brick, inverter, DC-DC converters), so no custom mains electronics are designed at portfolio level.
- DC bus kept below 60 V so the battery side stays in the extra-low-voltage range.
- Operating range about 0 to 40 °C ambient, with charging inhibited by the SwapCell BMS outside its limits.

■ Out of scope

- Connection to household wiring in any form, including transfer switches, double-male plug cords and grid-tie inverters.
- Loads above about 300 W, or with high starting surges: refrigerators, freezers, kettles, irons, hair dryers, space heaters, pumps and power tools.
- Whole-home backup and multi-day autonomy for large loads.

- Design of the SwapCell pack, the StepGen generator or the SunSpoke bike; PowerBox uses their interfaces.
- Design of new mains-voltage electronics. The inverter and grid charger are certified off-the-shelf units.

■ Prior work

- Commercial portable power stations from several brands (lithium iron phosphate or NMC, 250 to 2,000 Wh) show the product form and output mix users expect: USB-C PD, USB-A, 12 V car socket and pure sine AC. Almost all have sealed, non-swappable batteries.
- Pay-as-you-go solar home systems, widely deployed in East and West Africa and South Asia, show that 20 to 100 Wh per day of lights, phone charging and radio transforms evening life, and that honest sizing matters.
- Battery-swap networks for two- and three-wheelers show that a shared, standard pack can be charged centrally and carried home.
- Pedal and stepper generators have a long history in education and off-grid settings; the realistic output of a person over an hour is well under 100 W for most users, which sets expectations for human-powered charging.
- Open-source battery management and power electronics projects (for example open BMS designs used by the e-bike and solar DIY communities) provide starting points for the host controller and CAN handling.

No link is given where a specific reference has not been verified.

■ Open questions

- Which users to involve first, and through which partner (an NGO running community charging points, a solar home system distributor, or a university group in an outage-prone city)? Proposed, awaiting Amish.
- Which region to design the AC outlet for first (230 V 50 Hz or 120 V 60 Hz), which also fixes the plug type and GFCI or RCD standard? Proposed: 230 V 50 Hz with an RCD, awaiting Amish.
- Is theft resistance (a lockable pack door) a priority for informal settlement and community charging users?

■ User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [] Run co-design sessions with intended users; record who, where and what was learned
- [] Validate load, distance, terrain and cost assumptions in the field
- [] Revise requirements (REQ) from findings before freezing the design